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BRITISH STANDARD

BS EN ISO 13565-1:1998

Incorporating Amendment No. 1 to BS ISO 13565-1:1996 (renumbers BS as BS EN ISO 13565-1: 1998)

Geometric Product Specifications (GPS) — Surface texture: Profile method — Surfaces having stratified functional properties —

Part 1: Filtering and general measurement conditions

The European Standard EN ISO 13565-1:1997 has the status of a British Standard

ICS 17.040.20

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National foreword

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The UK participation in its preparation was entrusted to Technical Committee MTE/27, Properties and metrology of surfaces, which has the responsibility to:

— aid enquirers to understand the text;

— present to the responsible international/European committee any enquiries on the interpretation, or proposals for change, and keep the UK interests informed;

— monitor related international and European developments and promulgate them in the UK.

A list of organizations represented on this committee can be obtained on request to its secretary.

Cross-references

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Summary of pages

This document comprises a front cover, an inside front cover, pages i and ii, the EN title page, page 2, the ISO title page, pages ii to iv, pages 1 to 5 and a back cover.

This standard has been updated (see copyright date) and may have had amendments incorporated. This will be indicated in the amendment table on the inside front cover.

This British Standard, having been prepared under the direction of the Engineering Sector Board, was published under the authority of the Standards Board and comes into effect on 15 May 1997

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Amendments issued since publication

Amd. No.	Date	Comments
9949	May 1998	Indicated by a sideline in the margin

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EUROPEAN STANDARD

NORME EUROPÉENNE

EUROPÄISCHE NORM

EN ISO 13565-1

December 1997

ICS 17.040.20

Descriptors: See ISO document

English version

Geometrical product specifications (GPS) — Surface texture: Profile method; surfaces having stratified functional properties
Part 1: Filtering and general measurement conditions

(ISO 13565-1:1996)

Spécification géométrique des produits (GPS) — Etat de surface: Méthode du profil; surfaces ayant des propriétés fonctionnelles différentes suivant les niveaux
Partie 1: Filtrage et conditions générales de mesurage
(ISO 13565-1:1996)

Geometrische Produktspezifikzationen (GPS) — Oberflächenbeschaffenheit: Tastschnittverfahren; Oberflächen mit plateauartigen funktionsrelevanten Eigenschaften — Teil 1: Filterung und allgemeine Meßbedingungen
(ISO 13565-1:1996)

This European Standard was approved by CEN on 2 November 1997.

CEN members are bound to comply with the CEN/CENELEC Internal Regulations which stipulate the conditions for giving this European Standard the status of a national standard without any alteration. Up-to-date lists and bibliographical references concerning such national standards may be obtained on application to the Central Secretariat or to any CEN member.

This European Standard exists in three official versions (English, French, German). A version in any other language made by translation under the responsibility of a CEN member into its own language and notified to the Central Secretariat has the same status as the official versions.

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Ref. No. EN ISO 13565-1:1997 E

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Foreword

The text of the International Standard from Technical Committee ISO/TC 57, Metrology and properties of surfaces of the International Organization for Standardization (ISO) has been taken over as a European Standard by Technical Committee CEN/TC 290 Dimensional and geometrical product specification and verification, the secretariat of which is held by DIN.

This European Standard shall be given the status of a national standard, either by publication of an identical text or by endorsement, at the latest by June 1998, and conflicting national standards shall be withdrawn at the latest by June 1998.

According to CEN/CENELEC Internal Regulations, the national standards organizations of the following countries are bound to implement this European Standard: Austria, Belgium, Czech Republic, Denmark, Finland, France, Germany, Greece, Iceland, Ireland, Italy, Luxembourg, Netherlands, Norway, Portugal, Spain, Sweden, Switzerland and the United Kingdom.

Endorsement notice

The text of the International Standard ISO 13565-1:1996 has been approved by CEN as a European Standard without any modification.

NOTE Normative references to International Standards are listed in Annex ZA (normative).

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INTERNATIONAL STANDARD

ISO 13565-1

First edition
1996-12-01

Geometrical Product Specifications (GPS) — Surface texture: Profile method; Surfaces having stratified functional properties —

Part 1:
Filtering and general measurement conditions

Spécification géométrique des produits (GPS) — État de surface: Méthode du profil; surfaces ayant des propriétés fonctionnelles différentes suivant les niveaux —

Partie 1: Filtrage et conditions générales de mesurage

Reference number
ISO 13565:1996(E)

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Foreword

ISO (the International Organization for Standardization) is a worldwide federation of national standards bodies (ISO member bodies). The work of preparing International Standards is normally carried out through ISO technical committees. Each member body interested in a subject for which a technical committee has been established has the right to be represented on that committee. International organizations, governmental and non-governmental, in liaison with ISO, also take part in the work. ISO collaborates closely with the International Electrotechnical Commission (IEC) on all matters of electrotechnical standardization.

Draft International Standards adopted by the technical committees are circulated to the member bodies for voting. Publication as an International Standard requires approval by at least 75 % of the member bodies casting a vote.

International Standard ISO 13565-1 was prepared jointly by Technical Committees ISO/TC 57, *Metrology and properties of surfaces*, Sub-committee SC 1, *Geometrical parameters — Instruments and procedures for measurement of surface roughness and waviness,* ISO/TC 3, *Limits and fits*, and ISO/TC 10, *Technical drawings, product definition and related documentation*, Subcommittee SC 5, *Dimensioning and tolerancing*.

ISO 13565 consists of the following parts, under the general title *Geometrical product specifications (GPS) — Surface texture: Profile method; Surfaces having stratified functional properties*:

— *Part 1: Filtering and general measurement conditions;*

— *Part 2: Height characterization using the linear material ratio curve;*

— *Part 3: Height characterization using the material probability curve.*

Annex A and Annex B of this part of ISO 13565 are for information only.

Descriptors: Surface properties, surface condition, determination, characteristics, texture profiles, roughness measurement, general conditions, geometrical product specifications.

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Introduction

This part of ISO 13565 is a Geometrical Product Specification (GPS) standard and is to be regarded as a *General GPS standard* (see ISO/TR 14638). It influences chain links 2 and 3 of the chain of standards for roughness profile.

For more detailed information of the relation of this standard to other standards and the GPS matrix model, see Annex A.

The roughness profile generated using the filter defined in ISO 11562 suffers some undesirable distortions, when the measured surface consists of relatively deep valleys beneath a more finely finished plateau with minimal waviness. This type of surface is very common, for example in cylinder liners for internal combustion engines.

This part of ISO 13565 provides a method of greatly reducing these distortions, thus enabling the parameters defined in ISO 13565-2 and ISO 13565-3 to be used for evaluating the above mentioned type of surface, with minimal influence from these distortions.

1 Scope

This part of ISO 13565 describes a filtering method for use with surfaces that have deep valleys below a more finely finished plateau with a relatively small amount of waviness. The reference line resulting from filtering according to ISO 11562 for such surfaces is undesirably influenced by the presence of the valleys. The filtering approach described in this standard suppresses the valley influence on the reference line such that a more satisfactory reference line is generated.

2 Normative references

The following standards contain provisions which, through reference in this text, constitute provisions of this part of ISO 13565. At the time of publication, the editions indicated were valid. All standards are subject to revision, and parties to agreements based on this part of ISO 13565 are encouraged to investigate the possibility of applying the most recent editions of the standards indicated below. Members of IEC and ISO maintain registers of currently valid International Standards.

ISO 3274:1996, *Geometrical Product Specifications (GPS) — Surface texture: Profile method — Nominal characteristics of contact (stylus) instruments.*

ISO 4287:1996, *Geometrical Product Specifications (GPS) — Surface texture: Profile method — Terms, definitions and surface texture parameters.*

ISO 11562:1996, *Geometrical Product Specifications (GPS) — Surface texture: Profile method — Metrological characteristics of phase correct filters.*

3 Definitions

For the purposes of this part of ISO 13565, the definitions given in ISO 3274 and ISO 4287 apply.

4 Reference guide

To measure profiles in accordance with this part of ISO 13565, a measuring system which incorporates an external reference is recommended. In case of arbitration the use of such a system is obligatory.

5 Traversing direction

The traversing direction shall be perpendicular to the direction of lay unless otherwise indicated.

6 Filtering process to determine the roughness profile

The filtering process is carried out in several stages giving the modified profiles, sections of which are illustrated in Figure 1 a) to Figure 1 d).

The first mean line is determined by a preliminary filtering of the primary profile with the phase correct filter in accordance with ISO 11562 using a cut-off wavelength λc in accordance with clause **7** and corresponding measuring conditions in accordance with Table 1 of ISO 3274:1996. All valley portions which lie below this mean line [shown hatched in Figure 1 a)] are removed. In these places the primary profile is replaced by the curve of the mean line.

The same filter is used again on this profile with the valleys suppressed. The second mean line thus obtained [see Figure 1 b)] is the reference line relative to which the assessment of profile parameters is performed. This reference line is tranferred to the original primary profile [see Figure 1 c)] and the roughness profile according to this part of ISO 13565 is obtained from the difference between the primary profile and the reference line [see Figure 1 d)].

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7 Selection of the cut-off wavelength λc and the evaluation length *ln*

The measurements shall preferably be carried out using a cut-off wavelength of λc = 0,8 mm. In justified exceptional cases, λc = 2,5 mm may be selected and this shall be stated in the specification and test results.

Table 1 provides the relationship between cut-off wavelength λc and the evaluation length *ln*

Table 1 — Relationship between the cut-off wavelength λc and the evaluation length *ln*

Dimensions in millimetres

λc	*ln*
0,8	4
2,5	12,5

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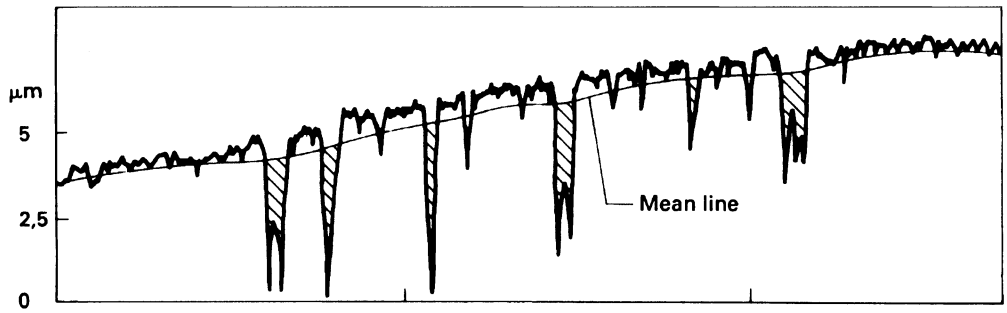
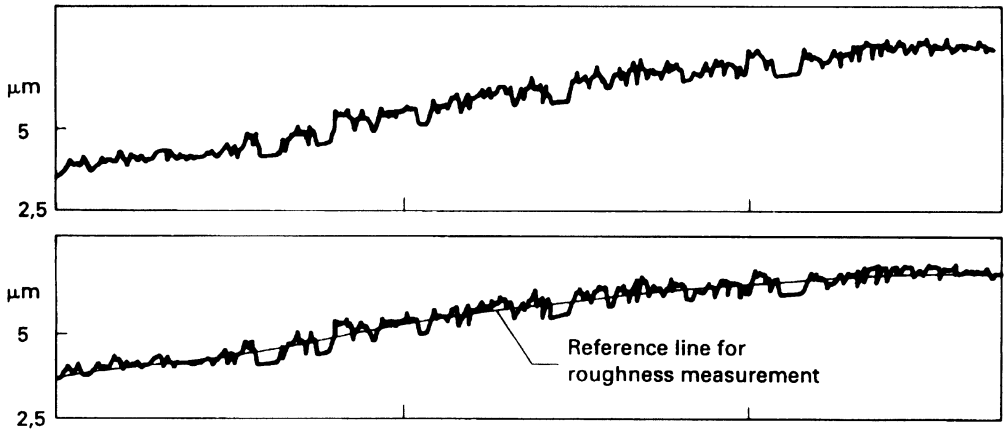
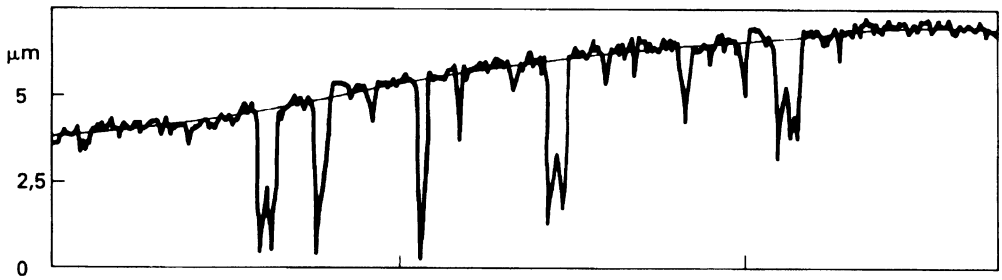


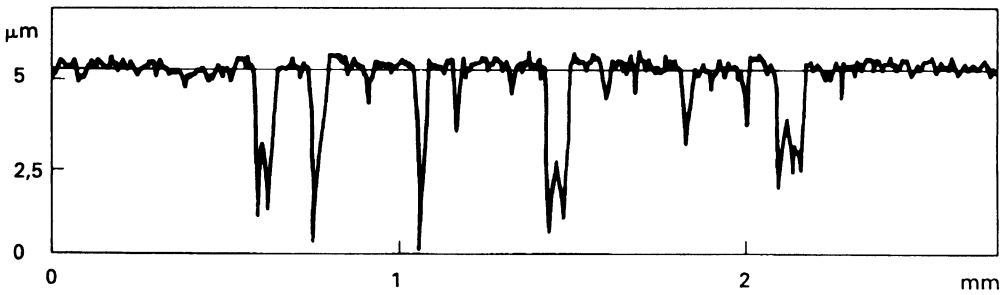
a) Unfiltered primary profile (valleys shown hatched)



b) Unfiltered primary profile after suppression of valleys



c) Position of the reference line in the primary profile



d) Roughness profile in accordance with this standard

Figure 1 — Filtering process

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Annex A (informative) Relation to GPS matrix model

For full details about the GPS matrix model see ISO/TR 14638.

A.1 Information about this part of ISO 13565 and its use

This part of ISO 13565 describes a filtering method for use with surfaces that have deep valleys below a more finely finished plateau with a relatively small amount of waviness. The roughness profile generated using the filter defined in ISO 11562 suffers some undesirable distortions, when measuring this type of surface, which is very common.

This method serves to reduce these distortions, thus enabling the parameters defined in ISO 13565-2 and ISO 13565-3 to be used for evaluating the above-mentioned type of surface, with minimal influence from these distortions.

A.2 Position in the GPS matrix model

This part of ISO 13565 is a *General GPS standard*, which influences the chain links 2 and 3 of the chain of standards for roughness profile in the *General GPS matrix*, as graphically illustrated in Figure A.1.

A.3 Related standards

The related International Standards are those of the chains of standards indicated in Figure A.1.

Fundamental GPS standards

Global GPS standards

General GPS matrix

Chain link number	**1**	**2**	**3**	**4**	**5**	**6**
Size						
Distance						
Radius						
Angle						
Form of line independent of datum						
Form of line dependent on datum						
Form of surface independent of datum						
Form of surface dependent on datum						
Orientation						
Location						
Circular run-out						
Total run-out						
Datum planes						
Roughness profile						
Waviness profile						
Primary profile						
Surface defects						
Edges						

Figure A.1

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Annex B (informative)
Bibliography

[1] ISO 13565-2:1996, *Geometrical Product Specifications (GPS) — Surface texture: Profile method; Surfaces having stratified functional properties — Part 2: Height characterization using the linear material ratio curve*.

[2] ISO 13565-3:—, *Geometrical Product Specifications (GPS) — Surface texture: Profile method; Surfaces having stratified functional properties — Part 3: Height characterization using the material probability curve*[1].

[3] ISO/TR 14638:1995, *Geometrical Product Specifications (GPS) — Masterplan*.

[4] *VIM — International vocabulary of basic and general terms in metrology*. BIPM, IEC, IFCC, ISO, IUPAC, IUPAP, OIML, 2nd edition, 1993.

Annex ZA (normative)
Normative references to international publications with their relevant European publications

This European Standard incorporates by dated or undated reference, provisions from other publications. These normative references cited at the appropriate places in the text and the publications are listed hereafter. For dated references, subsequent amendments to or revisions of any of these publications apply to this European Standard only when incorporated in it by amendment or revision. For undated references the latest edition of the publication referred to applies.

Publication	Year	Title	EN	Year
ISO 11562	1996	*Geometrical product specifications (GPS) — Surface texture: Profile method — Metrological characteristics of phase correct filters*	EN ISO 11562	1997

[1] To be published.

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